

AI and Quantum Technologies

From general considerations to a reinforcement-learning designed quantum sensor

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Time and place to be changed

What is this talk about?

Artificial intelligence and quantum technologies are the two computational revolutions of our time, and they feed each other. The first part of this talk is a pedestrian tour of *why* they are linked. The second part is one concrete example from our own work: how a reinforcement-learning agent can design an entangled quantum sensor made of two electrons floating on liquid helium.

Take-home message

Machine learning is becoming the design and control layer of quantum devices, and quantum devices are becoming new tools for sensing, simulation and, eventually, computing.

Thanks to many, and to our sponsors

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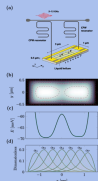
Outline

- 1 Two revolutions and why they meet
- 2 An example: a reinforcement-learning designed quantum sensor
- 3 Conclusions and perspectives

Quantum technology and machine learning/AI

Quantum Hardware

- Quantum control and sensors
- Quantum computers
- Catalysis



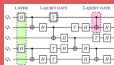
AI for Quantum Technology

- Quantum mechanical many-body theories
- Molecular dynamics
- Density functional theory

Quantum Technology and AI

Quantum Algorithms for AI

- Quantum computing
- Quantum algorithms
- Quantum machine learning
- Quantum inspired classical computing



Societal impact and products



New medicines



Better shipping



Financial innovation



Quantum readiness



Quantum sensors and computers

- Drug discovery
- Quantum sensing
- Supply chain optimization
- Financial market modelling and risk analysis
- Material discovery

What do we mean by AI and machine learning?

Here: a collection of methods that **learn a model from data by minimising a cost function**. At the heart of every machine-learning method there is an optimisation problem.

- **Supervised learning:** labelled data, regression and classification.
- **Unsupervised learning:** find structure without labels (clustering, dimensionality reduction, generative models).
- **Reinforcement learning (RL):** an *agent* acts on an *environment*, receives a *reward*, and learns a policy that maximises it.

$$\text{data } \mathbf{x} \rightarrow \text{model } f_{\theta} \rightarrow \text{cost } \mathcal{C}(\mathbf{x}, f_{\theta}) \rightarrow \min_{\theta} \mathcal{C}$$

The third family is the one we will use later: a physics simulator is the environment, electrode voltages are the actions, and the reward is a measure of sensor quality.

What do we mean by quantum technologies?

Devices whose function relies on **superposition**, **entanglement** and **interference** of well-controlled quantum states,

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle).$$

- ① **Quantum computing and simulation:** qubits and gates; Hilbert space grows as 2^N .
- ② **Quantum communication and cryptography:** security from physics, not from computational hardness.
- ③ **Quantum sensing and metrology:** entangled probes beat the shot-noise limit.

Platforms: superconducting circuits, trapped ions, neutral atoms, photons, spins in semiconductors, and, our favourite, **electrons on liquid helium**.

Quantum sensing in one slide

Estimate a parameter φ (a field, a phase, a frequency) encoded in a quantum state $|\psi(\varphi)\rangle$.

- N independent probes: standard quantum limit $\Delta\varphi \sim 1/\sqrt{N}$.
- N entangled probes: Heisenberg limit $\Delta\varphi \sim 1/N$.

The precision is bounded by the quantum Cramér–Rao bound and the quantum Fisher information F_Q ,

$$\text{Var}(\varphi_{\text{est}}) \geq \frac{1}{\nu F_Q(\varphi)},$$

with, for a pure state,

$$F_Q = 4 \left[\langle \partial_\varphi \psi | \partial_\varphi \psi \rangle \langle \partial_\varphi \psi | \partial_\varphi \psi \rangle - |\langle \psi | \partial_\varphi \psi \rangle|^2 \right].$$

Entanglement is a *resource*: it increases the variance of the generator that imprints φ , and hence F_Q . Differential (singlet) probes also reject common-mode noise.

Why do AI and quantum technologies need each other?

AI for quantum

- Device design and optimisation of control pulses and gate voltages
- Calibration, readout classification, error mitigation and decoding
- Neural-network quantum states for many-body simulation

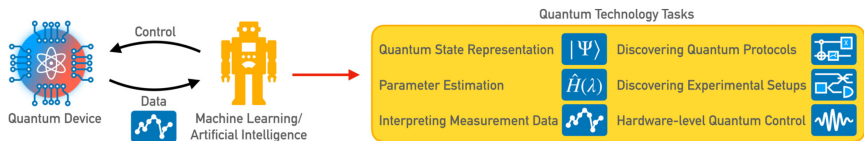
Quantum for AI

- Quantum machine learning: variational circuits, kernels
- Quantum simulators as data generators for hard physics
- Energy: can quantum hardware reduce the cost of some computations?

Common ground

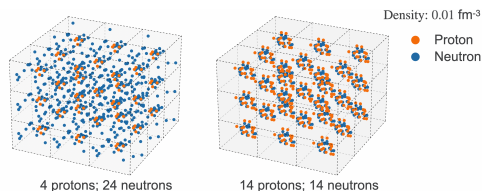
Both are optimisation problems in exponentially large spaces; both need physicists who speak both languages.

AI as a tool for quantum engineering



Machine learning enters at every stage: design of experiments, control, characterisation, and discovery of new protocols.

AI as a tool for quantum many-body physics

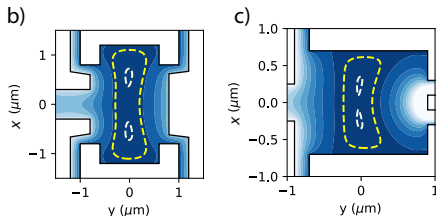
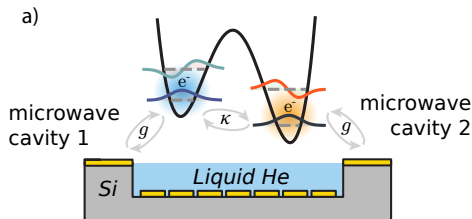


B. Fore, arXiv:2407.21207

- Neural-network quantum states solve the many-body Schrödinger equation variationally
- Self-emerging clustering in the crust of neutron stars
- Same ideas apply to electrons in quantum dots and on helium

Fore, Kim, Hjorth-Jensen, Lovato, Communications Physics **8**, 108 (2025).

Electrons on liquid helium as a quantum platform



- Electron floats in vacuum above a superfluid helium film: no lattice, no nuclear spins, no defects
- Predicted spin coherence times of seconds or more
- Electrodes below the helium shape a **double-well trap**; each electron couples to a microwave cavity
- Single-electron trapping and strong cavity coupling demonstrated experimentally

Panels (b),(c): gate geometries of Castoria et al. (2025) and Koolstra et al. (2026).

Theory: Beysengulov et al., PRX Quantum **5**, 030324 (2024); Leinonen et al. (2025).

The idea: an entangled pair as a differential sensor

Two electrons, one in each well, prepared in the spin singlet

$$|S\rangle = \frac{1}{\sqrt{2}} (|\uparrow_L \downarrow_R\rangle - |\downarrow_L \uparrow_R\rangle).$$

- A **uniform** field shifts both spins equally: the singlet does not care.
- A **difference** $\delta B = B_L - B_R$ between the wells couples the singlet to the triplet $|T_0\rangle$: the state starts to oscillate.
- The pair is a noise-cancelling magnetometer, sensitive to a localised perturbation, for instance from a passing particle or a weak exotic field.

Context

Proposed in line with CERN's Detector R&D initiative on quantum sensors (DRD5): rare, sub-keV events, ultralight dark-matter candidates, transient fields.

The physics model, in two equations

Two interacting electrons in a two-dimensional trap ($\hbar^2/2m_e = 1$),

$$\hat{H} = \sum_{i=1}^2 \left(-\frac{1}{2} \nabla_i^2 + v(\mathbf{r}_i) \right) + \frac{\kappa}{\sqrt{|\mathbf{r}_1 - \mathbf{r}_2|^2 + \epsilon^2}}.$$

- $v(\mathbf{r})$: electrostatic potential set by four electrode voltages
 $V = \{V_{\text{unload}}, V_{\text{trap}}, V_{\text{res}}, V_{\text{barrier}}\}.$
- One-body problem on a discrete-variable-representation grid; localised orbitals L0, L1, ... and R0, R1, ... by projecting on each well.
- Two-body problem by **full configuration interaction**: exact singlet and triplet states in the truncated basis.

A magnetic-field gradient enters through

$$H_B = g\mu_B \bar{B} (S_L^z + S_R^z) + \frac{g\mu_B \delta B}{2} (S_L^z - S_R^z), \quad \langle T_0 | H_B | S \rangle = \frac{g\mu_B \delta B}{2}.$$

How the signal is read out and how good it can be

Prepare the singlet-like ground state $|g\rangle$, wait a time t , measure the triplet-like population:

$$P_{g \rightarrow e}(t) = \frac{4\lambda^2}{\Delta_{ST}^2 + 4\lambda^2} \sin^2 \left[\frac{t}{2\hbar} \sqrt{\Delta_{ST}^2 + 4\lambda^2} \right], \quad \lambda = \frac{g\mu_B \delta B}{2} \mathcal{M}_{ge}.$$

- Δ_{ST} : singlet–triplet splitting; $\mathcal{M}_{ge} = \langle e | S_L^z - S_R^z | g \rangle$, equal to 1 for a perfect L0–R0 pair.
- Quantum Fisher information at short times:
 $F_Q^{ST} \simeq |\mathcal{M}_{ge}|^2 (g\mu_B t / \hbar)^2$.
- Two *independent* spins (Ramsey benchmark): $F_Q^{\text{ind}} = \frac{1}{2} (g\mu_B t / \hbar)^2$.

The ideal entangled pair gives **twice** the Fisher information of two independent probes. Everything that degrades the double well (leakage, delocalised orbitals, charge admixture) shrinks $|\mathcal{M}_{ge}|^2$.

Where AI enters: let an agent design the trap

The RL problem

- **Environment:** DVR + CI solver, time evolution under δB , Fisher information
- **Action:** the four electrode voltages
- **Reward:** sensor quality
- **Agent:** actor-critic network trained with Proximal Policy Optimisation (PPO), one-shot episodes

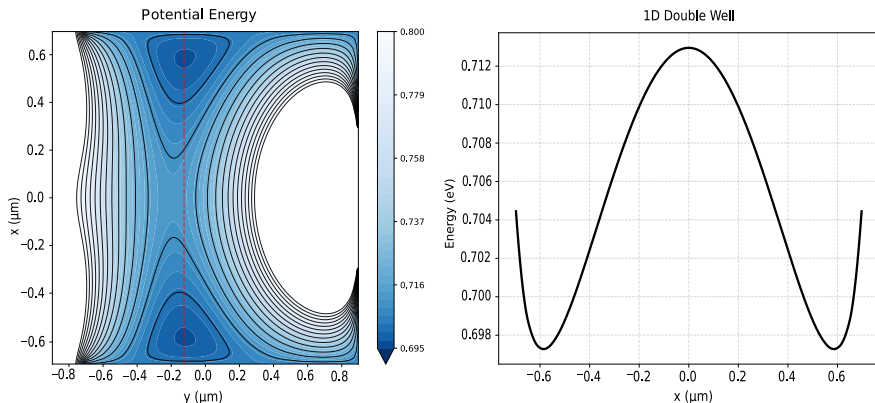
The reward

$$R(\mathbf{a}) = r_f r_{\text{cfi}} - (\omega_\mu p_\mu + \omega_b p_b)$$

- $r_f = F_{\text{singlet}} F_{\text{triplet}}$: fidelities of ground and first excited state
- r_{cfi} : classical Fisher information of the two- and three-channel readouts
- penalties for delocalised orbitals and for a barrier that is too low

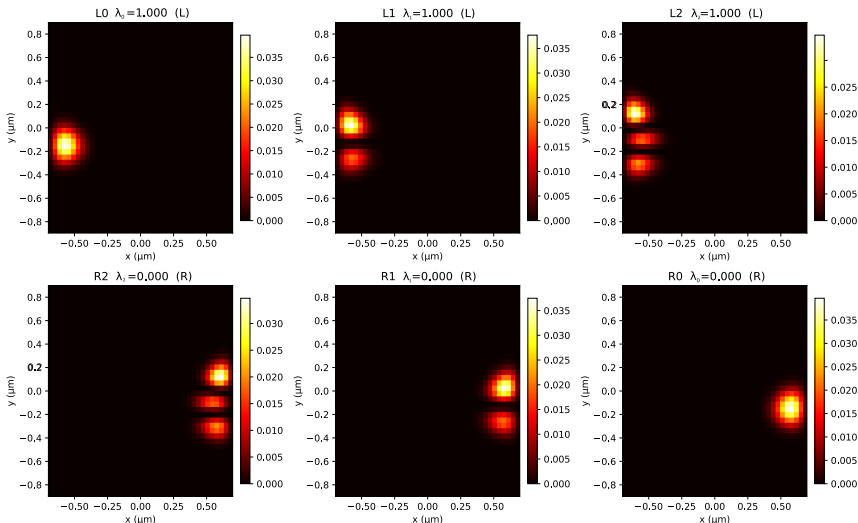
PPO: Schulman et al. (2017); RL for metrology: Belliardo et al. (2024), Bukov et al. (2026).

What the agent found: a clean double well



Left: two-dimensional potential (eV) for the RL-optimised voltages. Right: cut along the red dashed line ($y \approx -0.125 \mu\text{m}$). Wells about $1.5 \mu\text{m}$ apart, level spacings of 5–15 GHz.

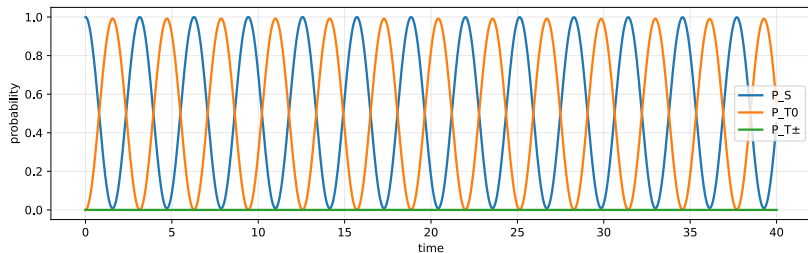
... with well-localised orbitals



One-body probabilities of the three lowest left-localised (L0, L1, L2, top) and right-localised (R0, R1, R2, bottom) orbitals. Localisation is what makes

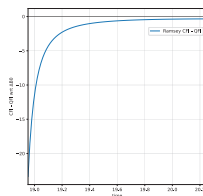
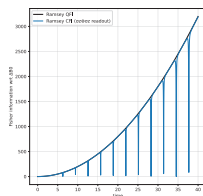
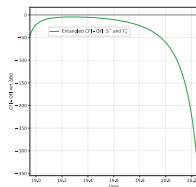
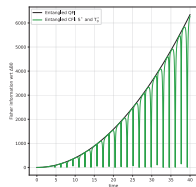
$|00\rangle = |0\rangle_L \otimes |0\rangle_R$ a good qubit pair.

Sensing dynamics: the two-level picture holds



Populations of $|S^*\rangle$, $|T_0^*\rangle$ and $|T_{\pm}^*\rangle$ under a field gradient for the RL-optimised trap. The full CI dynamics is well described by the effective singlet-triplet Hamiltonian. Time in units of μm^2 ($\hbar = 1$).

Fisher information: entangled probe versus independent spins

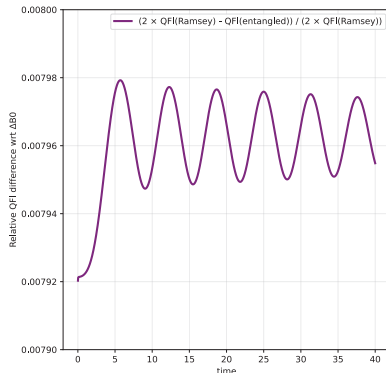
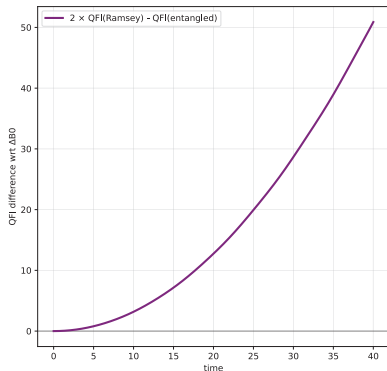


Entangled probe: quantum (QFI) and classical (CFI) Fisher information.

- The classical Fisher information saturates the quantum bound except at interrogation times where $\partial_{\delta B} P_{g \rightarrow e} = 0$ (the dips).
- In the optimal regime the entangled protocol reaches twice the Ramsey benchmark, within 1%.

Independent-spin Ramsey benchmark: half the Fisher information.

The factor of two is robust in time



Absolute (left) and relative (right) difference between $2 \times \text{QFI}(\text{Ramsey})$ and $\text{QFI}(\text{entangled})$. The relative difference stays constant within an envelope. Static field gradient only; time-dependent fields are future work.

From two qubits to a sensor network, and the hard parts

Scaling up

- Arrays of coupled electrons: collective response, $F_Q \propto N^2$
- Singlets for gradients (local events), GHZ-type states for common-mode fields (oscillating backgrounds)
- Same RL machinery can design the array

Challenges

- Millikelvin cryogenics and superfluid helium handling
- Single-electron spin readout: spin-to-charge conversion, quantum-limited amplifiers
- Fabrication uniformity across many traps
- Coupling the signal in without letting the noise in

This is a theoretical proposal: the physics is solved exactly for two electrons, the experiments so far characterise single electrons.

Observations (or conclusions if you prefer)

- Quantum sensing is where quantum technologies may pay off first, and entanglement gives a quantifiable gain: here a factor of two in Fisher information for a two-electron probe.
- Machine learning is the natural **design and control layer**: a reinforcement-learning agent, rewarded with a metrological figure of merit, found a trap geometry with clean, localised qubits.
- The recipe is general: physics simulator as environment, hardware knobs as actions, Fisher information as reward.
- Next: time-dependent signals, realistic noise, larger arrays, and closing the loop with experiments on electrons on helium.
- We need to train people who are fluent in both AI and quantum physics.

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- ④ C. L. Degen, F. Reinhard, P. Cappellaro, **Quantum sensing**, Rev. Mod. Phys. **89**, 035002 (2017)
- ⑤ CERN DRD5/RDq proto-collaboration, **Proposal on R&D on quantum sensors** (2024)

Thank you

Backup: PPO hyperparameters

Actor / critic optimiser	Adam (3×10^{-4} / 1×10^{-4})
Discount factor γ (one-shot)	0.0
GAE parameter λ	0.95
Clip range ϵ	0.2
Entropy coefficient	0.04
Critic loss weight	0.5
Max gradient norm	0.5
Rollout steps / minibatch	512 / 64
Update epochs	8

Backup: localised orbitals and natural orbitals

- Diagonalise the left-well projector $P_L = \int_{x < x_b} dx |x\rangle \langle x|$ in the low-energy single-particle space; eigenvalue $\lambda_\alpha \simeq 1$ (0) labels left (right) orbitals.
- Rank by $\tilde{\epsilon}_\alpha = \langle \phi_\alpha | H_{\text{sp}} | \phi_\alpha \rangle$ to get L0, L1, ..., R0, R1, ...
- From the CI state build the spin-summed one-body density matrix $\rho_{\mu\nu}^{(1)}$; its dominant eigenvectors in the L and R blocks define the reference singlet

$$|S^*\rangle = \frac{1}{\sqrt{2}} \left(d_{L\uparrow}^\dagger d_{R\downarrow}^\dagger - d_{L\downarrow}^\dagger d_{R\uparrow}^\dagger \right) |0\rangle$$

and triplet $|T_0^*\rangle$; overlaps with the CI eigenstates measure how ideal the qubit pair is.